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Quiz

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Main notions learned in this chapter

5/6 points (graded)

People find that getting ten "tails" in a row is (subjectively) less probable than getting tail-head-tail-head-tail-head-tail-head. Why?

☐ Because ten "tails" is causally harder to generate.

☒ Because ten "tails" is simpler to describe.



Answer

Correct: Because ten "tails" is simpler to describe.

Cognitive complexity is always smaller than plain complexity.

☐ True

☒ False



Answer

Correct: False. It is always larger, because it is resource-bounded.

The interest of news decays with time as $2 \times \log_2(t)$, where t is the time interval since the news.

☒ True

☐ False



Answer

Incorrect:

False. Interest decays as $\log_2(t)$ (without factor 2, as time is one-dimensional).

Are rhyming verses appreciated because the identity of sounds spares bits of information in description complexity?

☒ Yes

☐ No



Answer

Correct: Yes. It introduces unexpected simplicity.

When asked whether they are happy to find a folded 10-euro bill on the ground, people declare that they would be happier if it happened in the forest rather than on a commercial street in the evening. Why?

☒ Because causal complexity is larger in the forest story.

☐ Because the 10-euro banknote is simpler to describe if found in the forest.



Answer

Correct:

Because the causal complexity that brings the banknote there is larger in the forest than in the shopping street.

Why are important events celebrated more intensely after 10, 50 or 100 years?

- ☐ Because round anniversaries are improbable.
- ☒ Because round numbers are simple and diminish the description complexity of the event.
- ☐ Because round numbers are simple and diminish complexity drop.



Answer

Correct:

Because round numbers are simple and diminish the description complexity of the event.

Submit

You have used 1 of 1 attempt

* Partially correct (5/6 points)

Problem 1

0/2 points (graded)

In your region, one cat out of 10 000 is absolutely white. Let's write C_d ('white') the complexity of the concept 'white' out of context (since the word 'white' is among the 200 most frequent words, C_d ('white') $\approx$ 8 bits). How unexpected (in bits) would an all-white cat be, as compared with a standard cat?

- ☐ unexpectedness: $14 + C_d$ ('white')
- ☒ unexpectedness: $14 - C_d$ ('white')
- ☐ unexpectedness: $14 + C_d$ ('cat') + C_d ('white')
- ☐ unexpectedness: $14 + C_d$ ('cat') - C_d ('white')
- ☐ unexpectedness: $14 - C_d$ ('cat') + C_d ('white')
- ☒ unexpectedness: $14 - C_d$ ('cat') - C_d ('white')



Answer

Incorrect:

The complexity of selecting one all-white cat among 10 000 cats requires a causal complexity of $C_w(wc) \approx \log_2(10000) = 14$ bits.

Description complexity involves the notion of 'white' and, perhaps the notion of 'cat'.

So unexpectedness reads: $U(wc) \leq 14 - C_d$ ('cat') - C_d ('white') where C_d ('cat') should be omitted if 'cat' already belongs to the context.

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Problem 2

0/2 points (graded)

You are designing a program that retrieves old events worth to tell. Event $E(t, f, a)$ occurred at time t back in the past and involved a friend f who performed action a . Suppose that $E(1, f, a)$ is considered worth telling by the program ($t = 1$ day), but $E(60, f, a)$ is not. Now f is mentioned in the context, independently from E . This makes f available for free for telling E . What is the condition on the complexity $C_d(f)$ of f for $E(60, f, a)$ to become worth telling?

☐ $C_d(f) \geq \log_{2+}(60) = 6$ bits

☐ $C_d(f) \geq 2 \times \log_{2+}(60) = 12$ bits

☒ $C_d(f) \leq \log_{2+}(60) = 6$ bits

☐ $C_d(f) \leq 2 \times \log_{2+}(60) = 12$ bits



Answer

Incorrect:

The complexity of events grows as $\log_2(t)$. So what we get from the mention of f should at least compensate for the older date: $C_d(f) > \log_{2+}(60) = 6$ bits.

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You have used 1 of 1 attempt

✖ Incorrect (0/2 points)

Problem 3

2/2 points (graded)

In a fortune wheel game, the player launches the wheel which will then stop at a definite location of the circle. There are N discrete, equally probable, locations, each with its own color. We suppose that all locations are equivalent (no winning sector). Two players launch the wheel successively. If they reach the same location, the event is definitely interesting. If they reach neighbouring locations, the event is still noticeable. Let's suppose that the landing positions m_1 and m_2 are different. What is the condition on N and on m_1 and m_2 for getting a noticeable event?

☐ $\log(N) \gg \log(m_1) + \log(m_2) + 1$

☐ $\log(N) \gg \log(m_1) + \log(m_2)$

☒ $\log(N) \gg \log(|m_1 - m_2|) + 1$

☐ $\log(N) \gg \log(|m_1 - m_2|)$



Answer

Correct:

The situation is noticeable if the description complexity of the second outcome is significantly smaller than $\log(N)$. If we describe m_2 by reference to m_1 instead of by its absolute position, we should have $\log(N) \gg 1 + \log(|m_1 - m_2|)$. The

er by its absolute position, it should have $\text{reg}(m_2) = 1 + \text{reg}(m_1 - m_2)$. The additional bit is necessary to describe the direction of m_2 from m_1 .

Submit

You have used 1 of 1 attempt

✔ Correct (2/2 points)

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